

Boletín de exercicios do tema:
Integración de funcións reais de unha variable

Materia: Cálculo

Curso 2023/2024

Exercicio 1 *Calcular as seguintes integrais:*

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| 1. $\int \frac{dx}{\sqrt{5x-2}}$ | 12. $\int \log(x) dx$ | 24. $\int \sec(x)^2 \sqrt{\tan(x)} dx$ |
| 2. $\int \frac{x^3 dx}{\sqrt{1+x^4}}$ | 13. $\int \frac{\arctan(x)}{1+x^2} dx$ | 25. $\int \cos(x) \operatorname{sen}(x)^3 dx$ |
| 3. $\int \frac{e^{1/x}}{x^2} dx$ | 14. $\int x \operatorname{sen} x dx$ | 26. $\int \operatorname{sen}(x)^2 dx$ |
| 4. $\int \frac{x dx}{\sqrt{x+1}}$ | 15. $\int x^2(3x^3+14)^3 dx$ | 27. $\int \frac{1}{\cos(x)} dx$ |
| 5. $\int x\sqrt{x-1} dx$ | 16. $\int \frac{17x}{\sqrt[3]{6x^2+8}} dx$ | 28. $\int \frac{1}{1+\operatorname{sen}(x)} dx$ |
| 6. $\int x^2 e^{x^3} dx$ | 17. $\int x^2 7^{x^3+5} dx$ | 29. $\int \frac{\operatorname{sen}(x)}{8+2\cos(x)+\cos^2(x)} dx$ |
| 7. $\int \sqrt[5]{5x+6} dx$ | 18. $\int \frac{1}{e^x+1} dx$ | 30. $\int \frac{6}{\cos(x)+2\operatorname{sen}(x)\cos(x)} dx$ |
| 8. $\int x(5x^2-3)^7 dx$ | 19. $\int \frac{1}{9a^x+4a^{-x}} dx$ | 31. $\int \frac{1}{x\sqrt{x}} dx$ |
| 9. $\int \frac{\log(2x)}{\log(4x)} \frac{dx}{x}$ | 20. $\int \frac{x-1}{x^2-x-2} dx$ | 32. $\int \frac{1}{x\sqrt{x+1}} dx$ |
| 10. $\int \frac{\operatorname{sen}(x)^3}{\sqrt{\cos(x)}} dx$ | 21. $\int \frac{x dx}{(x-1)(x+1)^2}$ | 33. $\int \frac{1}{(1+2x)^{2/3} - (1+2x)^{1/2}} dx$ |
| 11. $\int x \log(x) dx$ | 22. $\int \frac{2x^2+7x-1}{x^3+x^2-x-1} dx$ | 34. $\int \frac{x}{(1+x)^{1/2} - (1+x)^{1/3}} dx$ |
| 23. $\int \frac{3x^2+3x+1}{x^3+2x^2+2x+1} dx$ | | |